Project 04 : Password Cracking with JTR

Contents

Background

Install John the Ripper from the Fedora Repository and use it to crack a weak password of a newly created user.

Objectives

1. Start Fedora server virtual machine.
2. Connect your VM to the internet.
3. Download JTR from the Fedora repo.
4. Create a new user.
5. Set password to a weak password.
6. Use cat & grep to filter account from shadow file.
7. Run john to crack password.
8. Document steps for inclusion in a project report to be submitted for review and grading in Blackboard.

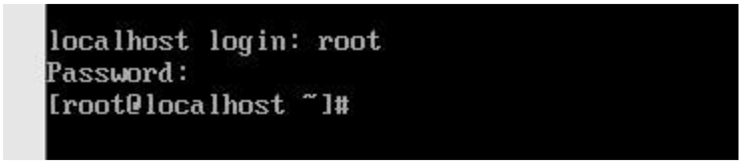
Prerequisites

1. A Windows or MacOS device with a functioning hypervisor (e.g., VMware Workstation Pro or VMware Fusion Pro).

Login

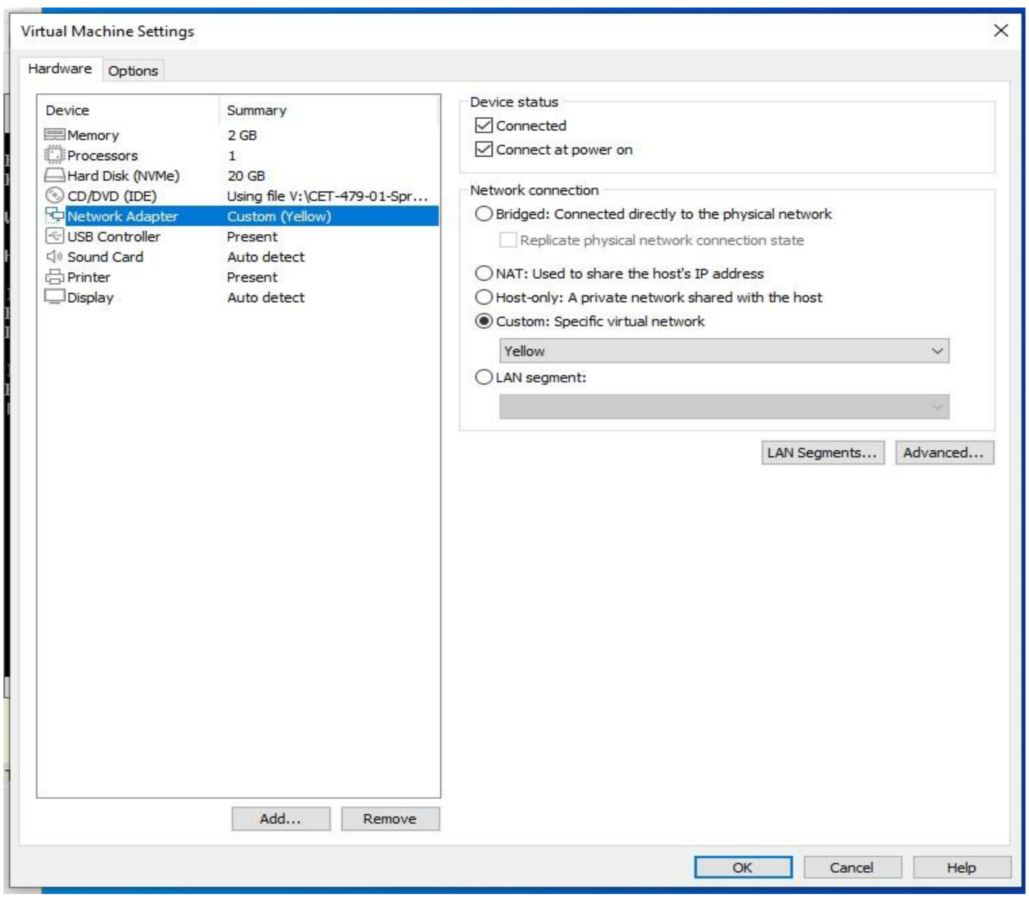
Connect your VM to the yellow network

1. Start your VM and log in as root.



2. Go to the VM option in the tab and click on settings.

3. Change the network adapter from NAT to custom and select the yellow option.



4. Type **`ifconfig`** . Notice that the IP address is not automatically assigned to your VM.

5. Type the following commands:

   # sudo nmcli connection modify **ens160 ipv4.method auto # sudo nmcli** connection down **ens160; sudo nmcli connection up ens160**

6. Type **ifconfig** again to check if the IP address was now assigned. Add screenshots.

```
[root@localhost ~]# sudo nmcli connection modify ens160 ipv4.method auto
[root@localhost ~]# sudo nmcli connection down ens160; sudo nmcli connection up ens160
Connection 'ens160' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/Act
iveConnection/2)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnecti
on/3)
[root@localhost ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.16.103  netmask 255.255.255.0  broadcast 192.168.16.255
        inet6 fe80::20c:29ff:fe69:8ed4  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:69:8e:d4  txqueuelen 1000  (Ethernet)
        RX packets 188  bytes 19498 (19.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 138  bytes 11979 (11.6 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[root@localhost ~]# _
```

Install John the Ripper Package and Create New User

7. Now type the following command **yum install john**

8. Create a new user named **jtrtest** and set the password to **abc123**. Show the commands below in a screenshot:

```
==================================================================================================
Installing:
 john              x86_64              1.8.0-21.fc37              fedora              8.9 M

Transaction Summary
==================================================================================================
Install  1 Package

Total download size: 8.9 M
Installed size: 20 M
Is this ok [y/N]: y
Downloading Packages:
john-1.8.0-21.fc37.x86_64.rpm                                    7.3 MB/s | 8.9 MB     00:01
--------------------------------------------------------------------------------------------------
Total                                                            6.8 MB/s | 8.9 MB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                        1/1
  Installing       : john-1.8.0-21.fc37.x86_64                                              1/1
  Running scriptlet: john-1.8.0-21.fc37.x86_64                                              1/1
  Verifying        : john-1.8.0-21.fc37.x86_64                                              1/1

Installed:
  john-1.8.0-21.fc37.x86_64

Complete!
[root@localhost ~]# useradd jtrtest
[root@localhost ~]# passwd jtrtest
Changing password for user jtrtest.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]#
```

Run john to crack the weak password

JTR contains a program named unshadow that combines the contents of the passwd and shadow files:

9. **cd ~**

10. **unshadow /etc/passwd /etc/shadow >mypasswd**

If we tried to crack the passwords in this file, it would take a long time due to the number of accounts contained in the file (all of them) so we want to isolate the account we just created.

11. Using the cat and grep commands, filter the user account jtrtest and create a new file named **cracklist** that contains only the single account jtrtest.

    a. **cat mypasswd|grep jtrtest >cracklist**

12. Show your command and display the contents of the file below:

13. Run the following command to crack the password of the jtrtest account:

john -format = crypt cracklist

Show your work

Once complete, you can use the following command to show the results:

14. **john – show cracklist**

```
[root@localhost ~]# cd ~
[root@localhost ~]# unshadow /etc/passwd /etc/shadow >mypasswd
[root@localhost ~]# cat mypasswd|grep jtrtest > cracklist
[root@localhost ~]# john -format=crypt cracklist
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
No password hashes left to crack (see FAQ)
[root@localhost ~]# john -show cracklist
jtrtest:abc123:1000:1000::/home/jtrtest:/bin/bash

1 password hash cracked, 0 left
[root@localhost ~]# _
```

Submission

Use this document (or one of your choosing) to submit your screen shots and any other relevant data. At a minimum, you should show the commands run after logging in as root. Please ensure your submission includes your name, class, date, and reference to this lab.